Solution Requirements

Date	30 june 2026
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (farmers, researchers, policymakers) access the OptiCrop web application directly without a login wall for the core crop analysis tool; an optional account/email-based sign-in is used for users who want to save analysis history.	Medium
2	Authorization levels	Three roles are supported: Farmer (enter soil/climate data, get crop recommendation), Researcher/Policymaker (view aggregated analytics and trends), and Admin (manage model versions and datasets). Each role accesses only the features relevant to it.	Medium
3	External interfaces	The system integrates with the trained Machine Learning models (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means) served via a Flask backend, and may connect to external weather/rainfall APIs for live climatic data.	High
4	Transactions processing	User-submitted soil and environmental parameters (N, P, K, temperature, humidity, pH, rainfall) are validated, sent to the Flask backend, processed through the ML pipeline, and the resulting crop recommendation is returned to the user in real time.	High
5	Reporting	The system generates a Crop Feasibility report showing the recommended crop, suitability score, and N-P-K/soil nutrition profiling insights; climate adaptive mapping output is also displayed as part of the result dashboard.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
6	Business rules, etc.	All input values (N, P, K, temperature, humidity, pH, rainfall) must be numerical and non-negative before the 'Optimize Production' action runs; the system selects the crop with the highest combined prediction confidence across the integrated models.	High
7	Compliance to laws or regulations	The system follows standard data privacy practices for any user-submitted soil/farm data and does not store personally identifiable information beyond what is required for analysis history.	Low
8	Other	Includes a 'Reset Fields' feature to clear inputs and a responsive mobile-first UI so farmers can use the tool directly from a phone in the field.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The crop optimization engine returns a recommendation almost immediately after the user submits soil and climate parameters, since the ML models run lightweight inference.	Result returned in < 3 seconds
2	Scalability	The Flask backend and ML inference layer are built to handle increasing numbers of simultaneous users (farmers, researchers) submitting analysis requests without performance degradation.	Supports up to 1,000 concurrent users
3	Security & Data Privacy	Soil and farm data submitted by users is transmitted securely and is not shared with third parties; only aggregated, anonymized data is used for research/policy analytics.	HTTPS/TLS encrypted data transfer
4	Reliability & Availability	The OptiCrop web application (hosted on Render) remains accessible to farmers and researchers with minimal downtime, including during peak usage periods such as planting season.	99.5% uptime per month
5	Usability & Accessibility	The interface uses clear icons, large touch-friendly input fields, and a mobile-first responsive layout so that farmers with limited technical literacy can easily enter data and read recommendations.	Mobile-responsive, WCAG 2.1 AA aligned
6	Other	Maintainability: the multi-algorithm ML pipeline (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means) is modular, allowing models to be retrained or swapped as new agricultural data becomes available.	Model retraining without downtime